

ESP32 BLUETOOTH JAMMER

Master Hardware Pinout & Wiring Specifications

LEGAL & EDUCATIONAL DISCLAIMER:

This device design and documentation are strictly for educational and testing purposes in controlled lab environments. RF signal jamming is illegal in many jurisdictions.

1. ESP32 Pin Allocation Overview

ESP32 30-PIN DEV BOARD LAYOUT

Radio 1 CE	GPIO 22	GPIO 21	Radio 1 CSN
Radio 2 CE	GPIO 16	GPIO 15	Radio 2 CSN
Touch T_CS	GPIO 27	GPIO 4	TFT DC / RS
TFT Backlight	GPIO 32	GPIO 5	TFT CS
TFT Reset	GPIO 33	GPIO 18	SPI SCK (Clock)
Physical ON Btn	GPIO 12	GPIO 19	SPI MISO
Physical OFF Btn	GPIO 14	GPIO 23	SPI MOSI
Power Rail	3.3V	GND	Ground Rail

2. Shared Hardware SPI Bus (VSPI)

Signal	ESP32 Pin	Connected Modules	Type
SCK (Clock)	GPIO 18	Radio 1, Radio 2, TFT SCK, Touch T_CLK	Shared SPI
MISO (Master In)	GPIO 19	Radio 1, Radio 2, TFT SDO, Touch T_DO	Shared SPI
MOSI (Master Out)	GPIO 23	Radio 1, Radio 2, TFT SDI, Touch T_DIN	Shared SPI

3. Radio Transceiver Modules (NRF24L01)

Radio #1 (Lower Channels 2402-2440 MHz):

NRF24 #1 Pin	ESP32 Pin	Function	Signal Category
VCC / GND	3.3V / GND	Power Rail (+3.3V Max)	Power / Ground
CE	GPIO 22	Chip Enable	Dedicated Control
CSN	GPIO 21	SPI Chip Select	Dedicated SPI CS
SCK / MOSI / MISO	GPIO 18 / 23 / 19	Hardware SPI Bus	Shared SPI

Radio #2 (Upper Channels 2441-2480 MHz):

NRF24 #2 Pin	ESP32 Pin	Function	Signal Category
VCC / GND	3.3V / GND	Power Rail (+3.3V Max)	Power / Ground
CE	GPIO 16	Chip Enable	Dedicated Control
CSN	GPIO 15	SPI Chip Select	Dedicated SPI CS
SCK / MOSI / MISO	GPIO 18 / 23 / 19	Hardware SPI Bus	Shared SPI

4. ILI9341 2.8" SPI Touchscreen Display & Touch Controller

Display / Touch Pin	ESP32 Pin	Function	Signal Category
VCC / GND	3.3V / GND	Display Power Rail	Power / Ground
CS	GPIO 5	TFT Chip Select	Dedicated SPI CS
RESET / RST	GPIO 33	TFT Hardware Reset	Dedicated Control
DC / RS	GPIO 4	Data / Command Select	Dedicated Control
LED / BL	GPIO 32	Screen Backlight PWM	Dedicated Control
SDI (MOSI) / SDO (MISO)	GPIO 23 / 19	TFT Display SPI Data	Shared SPI
SCK / CLK	GPIO 18	TFT Display SPI Clock	Shared SPI
T_CS	GPIO 27	Touch Controller CS	Dedicated SPI CS
T_CLK / T_DIN / T_DO	GPIO 18 / 23 / 19	Touch Controller SPI	Shared SPI

5. Hardware Pushbuttons & System Controls

Button / LED	ESP32 Pin	Wiring Connection	Mode
TURN ON Button	GPIO 12	One leg to GPIO 12, other leg to GND	INPUT_PULLUP
TURN OFF Button	GPIO 14	One leg to GPIO 14, other leg to GND	INPUT_PULLUP
BOOT Button	GPIO 0	Built-in BOOT pushbutton on ESP32 board	Internal
Status LED	GPIO 2	Built-in LED on ESP32 board	OUTPUT

6. Power Supply & Capacitor Recommendations

1. Built-in PCB Antenna Modules: Draw ~12mA each. Can be powered directly from ESP32 3.3V pin without capacitors.
2. High-Power PA+LNA Modules: Draw ~115mA peak each. Add a 10uF-47uF capacitor across VCC & GND of each module to prevent brownouts.